Managing Files and Directories

Commands

General knowledge

Options and Arguments:

- Commands are often followed by options that modify/enhance their behavior.
- Commands are also followed by arguments which are the items open which the command acts on.

`command -option argument` `ls -l ~/Downloads`

Command	Definition	Usage	Examples
mkdir	creates a single directory or multiple directories	'mkdir + the name of the directory'	'mkdir wallpapers', 'mkdir ~/wallpapers/forest'
touch	updates any given file's timestamp, if the file does not exist, it creates it.	'touch + name of the file'	'touch list', 'touch ~/Downloads/games.txt'
rm	removes files; can also remove directories	'rm + name of file', 'rm -r + directory name or directory absolute path'	'rm list', 'rm -r Downloads/games'
rmdir	removes empty directories	'rmdir + name of directory'	'rmdir Downloads/games'
mv	moves and renames directories	'mv + source + destination	'mv Downloads/homework.pdf Documents/'
cp	copies files/directories from a source to a destination	For files: 'cp + files to copy + destination', For directories: 'cp -r + directory to copy + destination'	'cp Downloads/wallpapers.zip Pictures/', 'cp -r ~/Downloads/wallpapers ~/Pictures/'

Command	Definition	Usage	Examples
ln	used to create hard links or soft links	For hard link: 'ln file ~/Downloads/fileHL', For soft link: 'ln -s file fileSL'	
man	used to access manual pages that are documentation files that describe Linux shell commands, executable programs, system calls, special files, etc...	'man + command'	'man passwd', 'man ls'

Wildcards

Using Wildcards / File Globbing (quick reference)

Wildcard	Description
*	Matches zero or more characters in a filename
?	Matches any one character in a filename
[acf]	Matches one of multiple characters in a filename; in this example, a, c, or f
[a-f]	Matches one of a range of characters in a filename; in this example, any character from a through f
[!a-f]	Matches filenames that don't contain a specified range of characters; in this example, filenames that don't contain a through f

Examples for * wildcard:

- To list files that end in .txt regardless of the size of the file.
  - ls *.txt
- To list files that end in .pdf and .png
  - ls *.pdf *.png

Examples for ? wildcard:

- To list all hidden files in the current directory
  - ls ./.??*
- To list files that have two characters between letter g and d
  - ls g??d*
- To list files that have a 4 letter file extension

        - ls *.???

Examples for [] wildcard:

- To match all files that have a range of letter after d:
    - 'ls d[a-z]*'

Brace Expansion

- Brace expansion {}
    - Is not a wildcard
    - Feature of bash that allows you to generate arbitrary strings to use with commands.
        - Examples:
            - To create a whole directory structure in a single command:
                - mkdir -p music/{rock,rap}/{mp3files,videos,oggfiles}/new{1..3}
            - To create N number of files:
                - touch website{1..5}.html
                - touch file{A..Z}.txt
                - touch file{001..10}.py
                - touch file{{a..z},{0..10}}.js
            - Remove multiple files in a single directory
                - rm -r {dir1,dir2,dir3,file.txt,file.py}